



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the Performance of Data Analysis Students

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Introduction

In an increasingly digital world, the way we handle money has shifted from physical bills to invisible data. This report explores how various payment methods ranging from traditional cash to modern e wallets influence the daily spending behavior and psychological pain of paying among consumers in Egypt.

Research Question

The impact of payment method (cash or digital) on daily expenses

Hypothesis

Users who primarily utilize digital payment methods will experience a higher frequency of emergent purchases and feel they spend more freely than those who primarily use physical cash.

Population of Interest:

The population of interest consists of individuals in Egypt, primarily within the 18 - 24 age group, who are active users of both physical and digital financial services.

Sampling Method:

A Convenience Sampling method was used by distributing an online survey via Google Forms. This was the most effective approach for this study as it targeted tech-savvy individuals who are most likely to use a variety of payment methods and are readily accessible through university and social networks.

Bias Identification:

Selection bias may occur because most respondents are students and young adults, which limits the diversity of the sample.

Response bias may also occur if participants do not accurately report their spending habits.

To reduce bias, the survey was kept anonymous to encourage honest responses. Additionally, the questions were written in a neutral and simple way to avoid influencing of participants answers.

Survey Questions:

- What is your age group?
- What's your Gender ?
- What is your primary method of payment for daily personal expenses?
- On average, how much money (in EGP) do you spend daily on non-essential items (e.g., coffee, snacks, entertainment)?

- How many times per week do you typically make an "Emergent purchase" (buying something you did not plan to buy)?
- How often do you actively track your daily spending?
- In your personal experience, do you feel you spend more freely when using digital payment methods compared to using physical cash?
- How many active paid monthly subscriptions do you currently have (e.g., Netflix, Spotify, Gym, Anghami)?
- What makes you choose to pay with physical CASH over digital methods?
- What makes you choose to pay with DIGITAL methods (Cards, E-Wallets) over cash?

Online survey link:

https://docs.google.com/forms/d/e/1FAIpQLSf1RPbHlerQyiiPQRhmDiOlzrTN0bqJoFtCtxEWxFM7_Nc2fg/viewform?usp=header

Number of samples collected: 51

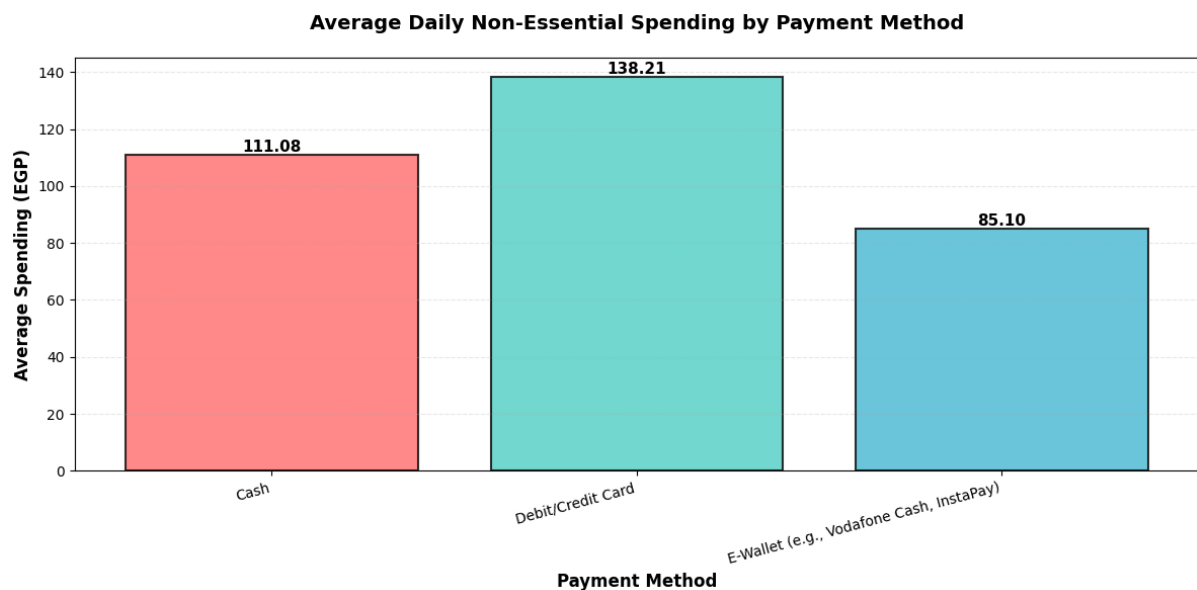
Analysis:

This chart compares average daily spending on non-essential items across three payment methods.

Debit/Credit Card users spend the most (138.21 EGP on average),

followed by Cash users (111.08 EGP), and E-Wallet users (85.10 EGP).

This suggests that digital payment methods (especially cards) may encourage higher spending on non-essential items.

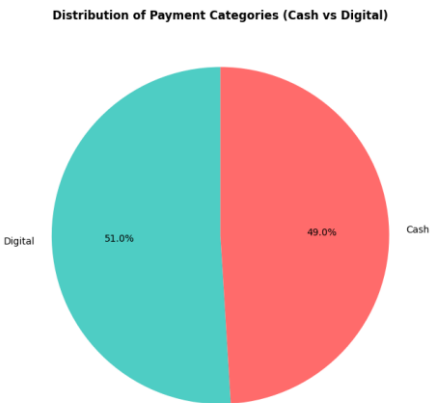


This chart visualizes the overall distribution of users between Cash and Digital payment categories.

The majority of users prefer Digital (51.0%),

while Cash accounts for the remaining 49.0%.

This indicates a strong shift towards digital adoption within this user base.



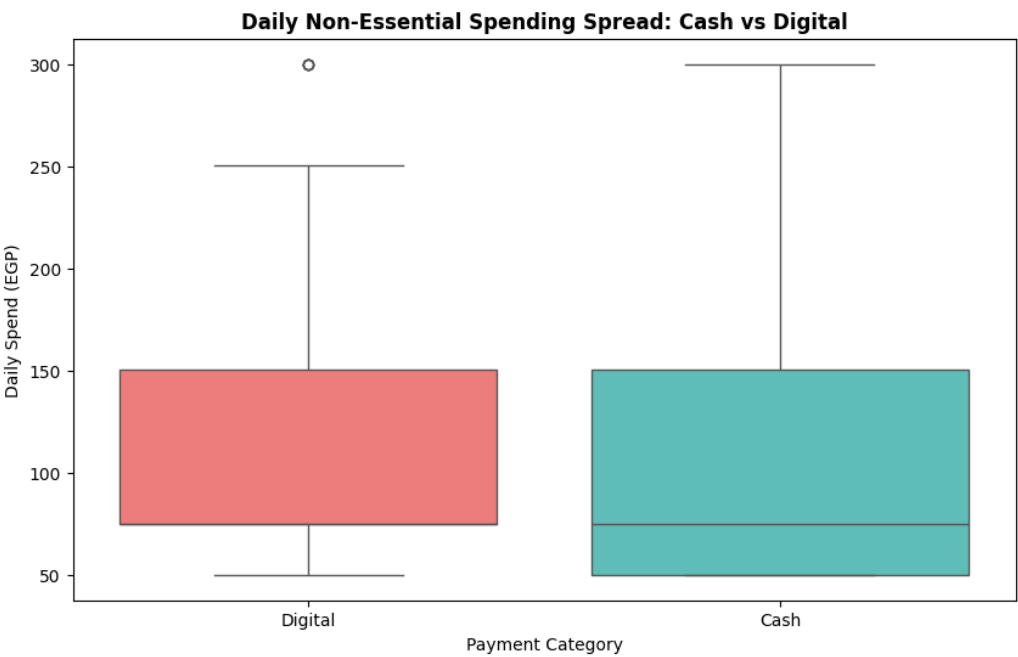
This box plot highlights the distribution, central tendency, and variability of daily non-essential spending.

The median daily spend for Cash users (75.00 EGP)

tends to be higher than that of Cash users (75.00 EGP).

Additionally, the Cash category shows a wider interquartile range (the size of the box).

This indicates greater variability in Cash spending habits, suggesting these users are more prone to occasional high-value impulse purchases, which often appear as outliers above the top whisker.



Conclusion

The results of this study support the hypothesis that payment methods influence spending behavior. Participants who use digital payment methods reported higher levels of spending compared to those who use cash.

Digital payments reduce the physical sensation of spending money, which may lead to less controlled financial behavior. However, they also provide convenience and efficiency.

Overall, while digital payment methods offer many advantages, they may also encourage increased spending if not used carefully.

Any potential issues

- **Sample Size and Homogeneity:** With a sample size of 51 respondents collected via convenience sampling, the dataset is relatively small and heavily skewed toward tech-savvy young adults (18–24). This limits the study's external validity, meaning the conclusions cannot be confidently generalized to the broader Egyptian population, which has varying levels of financial literacy and banking access.
- **Self-Reporting Inaccuracies:** The data relies entirely on participants estimating their own daily non-essential spending and frequency of emergent purchases. Financial tracking is notoriously prone to recall bias; individuals frequently underestimate their impulse buying or forget small daily transactions, meaning the reported figures may be lower than actual spending.
- **Missing Confounding Variables:** A major hidden bias is the absence of income or allowance data. It is highly likely that individuals with higher disposable incomes naturally gravitate toward using debit or credit cards. Therefore, the higher spending observed in card users might be a byproduct of their overall wealth rather than the payment method itself influencing their behavior.
- **Oversimplification of the "Digital" Category:** While the conclusion groups all digital methods together as encouraging increased spending, the data reveals a deeper nuance. E-wallet users actually reported the lowest average spending (85.10 EGP)—even lower than cash users. Grouping e-wallets and cards together masks the fact that pre-loaded digital wallets might actually promote budgeting, whereas the seamlessness (or credit line) of a physical card drives the higher spending.